On Estimation and Sensitivity Analysis of Distribution Circuit's Photovoltaic Hosting Capacity

Anamika Dubey, *Member, IEEE*, and Surya Santoso, *Fellow, IEEE*

***Abstract*—A high penetration of residential photovoltaic (PV) panels can potentially cause a number of operational issues in the distribution circuit, necessitating the requirement to determine the largest PV generation, a distribution circuit can accommodate. This paper presents a Monte Carlo-based hourly stochastic analysis framework to determine a circuit's PV accommodation limit or PV hosting capacity. First, a mathematical formulation of the hosting capacity problem for bus overvoltage concerns is presented, followed by the proposed analysis approach. In the proposed approach, the Monte Carlo method is used to simulate the scenarios of probable PV locations and sizes while the hourly analysis framework is used to include the daily variations in load and PV generation. Next, a method to evaluate the percentage accuracy of the hosting capacity results is presented. The sensitivity of PV hosting results to multiple parameters related to both feeder characteristics and simulation method are evaluated as well. The utility of the proposed framework is presented using an actual 12.47-kV distribution circuit. In sum, by presenting a PV hosting analysis method, accuracy assessment framework, and sensitivity analysis approach, this paper provides a thorough understanding of the PV hosting capacity problem.**

***Index Terms*—Distributed energy resources, photovoltaic systems, power quality, solar energy, voltage.**

I. Introduction

THE integration of distributed energy resources (DERs), especially residential photovoltaic (PV) systems, into the distribution circuit has rapidly increased in the past decade [1]. Since utility distribution systems are designed for centralized power generation and are optimized for unidirectional power flow, the integration of DERs may disrupt system's normal operating conditions. In literature, several studies have been conducted to determine the impacts of PV generation on distribution systems. A high PV penetration may result in several operational issues, including but not limited to voltage quality problems [2]–[12], increased thermal stress, additional feeder losses [13], and a higher number of capacitor switching and regulator tap operations [14]–[15].

The increasing PV penetration and the undesirable impacts of PVs on the distribution circuit necessitate the task of determining the largest PV capacity a given distribution circuit can accommodate without violating the circuit's operational limits. The obtained PV capacity is referred to as the circuit's PV hosting capacity or PV accommodation limit. The PV hosting capacity can be defined with respect to several impact criteria: for example, system overvoltage, thermal stress, harmonics, etc. However, given the strict regulations enforced for the bus overvoltage [16] condition, this paper defines and formulates feeder's PV hosting capacity problem for bus overvoltage concerns.

In the literature, several methods for determining the feeder's maximum PV penetration limit have been proposed [17]–[24]. However, the existing methods have a few limitations. A simplified feeder model was used in [18]–[20], thus not representing the actual circuit conditions. In [17]–[20], the PV systems were simulated only at a few specific locations, thus not considering the stochasticity of potential residential PV panel sizes and locations. In [18], [19], small test feeders were used, thus questioning the method's application for an actual distribution circuit. In [22]–[25], a stochastic analysis framework is used to obtain the PV hosting capacity of actual distribution feeders. The method simulates a large number of potential PV deployment scenarios by varying residential PV panel locations and sizes. However, the method poorly considers representative load condition for hosting capacity calculation. Additionally, the accuracy of the PV hosting results is not quantified, which is required given that the analysis framework is stochastic.

In this paper, we present a detailed analysis on formulating, solving, and understanding feeder's PV hosting capacity problem for overvoltage concerns. A mathematical formulation is laid out to define PV hosting capacity problem. An hourly stochastic analysis framework is proposed to calculate feeder's PV hosting capacity. The proposed framework includes the impacts of hourly variations in load demand and PV generation thus improving upon the state-of-the-art PV hosting methods. An accuracy assessment framework is also presented to quantify the accuracy of PV hosting capacities obtained using Monte Carlo simulations. Finally, sensitivity analyses are conducted to evaluate the impacts of distribution circuit characteristics and simulation parameters on feeders' PV hosting capacity.

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A. Dubey was with the University of Texas at Austin, Austin, TX 78712 USA, and is now with the School of Electrical Engineering and Computer Science, Washington State University, Pullman, WA 99164 USA (e-mail: anamika.dubey@wsu.edu).

S. Santoso is with the Department of Electrical and Computer Engineering, The University of Texas at Austin, Austin, TX 78712 USA (e-mail: ssantoso@mail.utexas.edu).

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Although this paper calculates PV hosting capacity for overvoltage concerns, the proposed approach is general and applicable to any other impact criteria.

A. Contributions

The specific contributions of this paper are highlighted in this section.

1) Mathematical Formulation: A mathematical formulation to define PV hosting capacity problem for overvoltage concerns is laid out. The proposed formulation provides insights and understanding of the complexity of PV hosting analysis. Moreover, the formulation can be extended to include any operational constraint for a given feeder.

2) Hourly Stochastic Analysis Framework: - A Monte Carlo based hourly stochastic analysis framework is developed to estimate the PV hosting capacity of a given feeder. The proposed framework includes the impacts of hourly variations in load demand and PV generation and results in a better estimate of the feeder's PV hosting capacity. The hourly analysis also identifies time periods of the day PV generation is likely to be of higher concern. The proposed approach, therefore, can be used for making operational planning decisions.

3) Accuracy Assessment Approach: An accuracy assessment framework is developed for determining the accuracy of the values calculated for feeder's PV hosting capacity. Since feeder's PV hosting capacity is estimated using Monte Carlo simulations while simulating only a small percentage of the possible scenarios, it is imperative to quantify the accuracy of the calculated values. The proposed approach facilitates in the evaluation and validation of the accuracy of the obtained PV hosting capacity results.

4) Sensitivity Analysis: Finally, we also evaluate the sensitivity of the feeder's PV hosting capacity with respect to multiple circuit characteristics and simulation parameters. The sensitivity analysis helps assess the relative impacts of multiple parameters on feeder's PV hosting capacity. The conclusions drawn can be used for a comparative study of a large number of secondary feeders in a limited amount of time. The study also helps quantify the number of Monte Carlo runs required to obtain sufficiently accurate hosting capacity estimates.

B. Paper Organization

The paper organization is as follows. Section II presents the mathematical formulation for the PV hosting capacity problem. Section III details the proposed Monte Carlo based hourly stochastic analysis framework, and accuracy assessment approach. The proposed method is implemented on a 12.47-kV utility distribution circuit, supplied by a 24-MVA substation transformer. The results and discussions are presented in Section IV, followed by conclusions in Section V.

II. PV Hosting Capacity

A feeder's PV hosting capacity is defined as the largest PV generation that can be accommodated without violating the feeder's operational limits. This paper is primarily focussed on the primary wire overvoltage conditions resulting from PV integration. An overvoltage violation is recorded if any primary bus in the feeder records a voltage greater than 1.05 pu. The hosting capacity problem aims to determine the largest PV generation that can be integrated into a given feeder without resulting in an overvoltage violation. Since, the locations and sizes of the future residential PV panels are not known, the hosting capacity problem requires simulating and analyzing a large number of potential PV deployments.

A. Definitions

This section defines the terms used in the mathematical formulation of the PV hosting capacity problem described in Section II-B.

1) Customer Penetration Level (C^i_{pen}): Customer penetration level defines the number of customers (in percentage) equipped with PV panels in a given distribution circuit. Thus, an ith customer penetration level is obtained by populating $i\%$ customers with PV panels. Here, i is a discrete variable, $i \in S$, where S is the set of discrete customer penetration levels.

2) PV Penetration Level (PV^i_{pen}): PV^i_{pen} is defined as the total PV generation in MW or kW added to the distribution circuit corresponding to the ith customer penetration level (C^i_{pen}). A PV penetration level (PV^i_{pen}) is obtained corresponding to each customer penetration level (C^i_{pen}).

3) PV Deployment Scenarios (x^i_j): A PV deployment scenario characterizes the locations and sizes of individual PV panels at each customer load location for a given customer penetration level. Multiple potential PV locations and sizes characterizing a PV deployment scenario corresponding to each customer penetration level are possible. Here, a PV deployment scenario, x^i_j, represents j^{th} PV deployment scenario corresponding to i^{th} customer penetration level.

B. PV Hosting Capacity - Problem Formulation

The results of a large number of potential PV deployment scenarios are quantified using two representative PV hosting capacities, minimum-hosting ($H_{\min}$) and maximum-hosting capacity ($H_{\max}$). The minimum-hosting capacity ($H_{\min}$) is defined as the lowest PV generation resulting in the first case of overvoltage condition. The maximum-hosting ($H_{\max}$) capacity is defined as the lowest PV generation corresponding to which all PV deployment scenarios report an overvoltage.

Mathematically, the PV hosting capacity problem for a given distribution circuit is defined as follows. Let,

S be the set of discrete customer penetration levels indexed by i, $S \in \{1, 2, \ldots, i, \ldots, 100\}$.

PV_{pen} be the set of all PV penetration levels indexed by customer penetration level, i, $\{PV^1_{\text{pen}}, PV^2_{\text{pen}}, PV^3_{\text{pen}}, \ldots, PV^i_{\text{pen}}, \ldots, PV^{100}_{\text{pen}}\}$.

X^i be the set of all PV deployment scenarios corresponding to C^i_{pen}, indexed by j, $\{x^i_1, x^i_2, \ldots, x^i_j, \ldots, x^i_{L_i}\}$.

L_i be the total number of PV deployment scenarios corresponding to C^i_{pen}.

$V^i_{\max}$ be the set of largest primary voltages recorded for all deployment scenarios corresponding to C^i_{pen}. $\{V_{\max}(x^i_1), V_{\max}(x^i_2), \ldots, V_{\max}(x^i_j), \ldots, V_{\max}(x^i_{L_i})\}$, whe

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re, $V_{\max}(x_j^i)$ is the largest primary voltage recorded for the PV deployment scenario x_j^i.

The hosting capacity problem is defined as follows. Determine $H_{\min}$ and $H_{\max}$, where:

$$H_{\min} = \min_{i \in S} \left\{ PV_{\text{pen}}^i | \max(V_{\max}^i) > 1.05 \right\} \quad (1)$$

$$H_{\max} = \min_{i \in S} \left\{ PV_{\text{pen}}^i | \min(V_{\max}^i) > 1.05 \right\} \quad (2)$$

C. Monte Carlo Based Method for PV Hosting Problem

The minimum- and maximum-hosting capacity as defined in (1) and (2) can be obtained only after simulating all possible PV deployment scenarios, which is impractical. Therefore, the PV hosting analysis is done by simulating a finite number of PV deployment scenarios using the Monte Carlo approach. The PV hosting problem based on the Monte Carlo approach is defined in this section.

For a given distribution circuit, let k be the number of PV deployment scenarios simulated at each customer penetration level, each representing one Monte Carlo run. The study resulting from k Monte Carlo runs is termed k-*run* Monte Carlo study (k-*run* MCS). For a k-*run* MCS, the feeder's hosting capacity is characterized using the first- and all-hosting capacity defined as follows:

1) First-Hosting Capacity ($H_{1,k}$): The first-hosting capacity is equal to the lowest PV penetration for which at least one scenario in a k-*run* MCS observes an overvoltage condition.

2) All-Hosting capacity ($H_{100,k}$): The all-hosting capacity is defined as the lowest PV penetration such that all k scenarios in a k-*run* MCS observe an overvoltage condition.

$H_{1,k}$ and $H_{100,k}$ are obtained by solving (3) and (4).

$$H_{1,k} = \min_{i \in S} \left\{ PV_{\text{pen}}^i | P(V_{\max,k}^i > 1.05) \geq \frac{1}{k} \right\} \quad (3)$$

$$H_{100,k} = \min_{i \in S} \left\{ PV_{\text{pen}}^i | P(V_{\max,k}^i > 1.05) = 1 \right\} \quad (4)$$

where,

X_k^i: Set of k PV deployment scenarios simulated using Monte Carlo method at C_{pen}^i, $\{x_1^i, x_2^i, x_3^i, \ldots, x_k^i\}$.

$V_{\max,k}^i$: Set of maximum primary voltages recorded for k PV deployment scenarios simulated at C_{pen}^i, $\{V_{\max}(x_1^i), V_{\max}(x_2^i), \ldots, V_{\max}(x_k^i)\}$.

III. Proposed Methodology

This section details the proposed stochastic analysis framework to solve the PV hosting capacity problem. A novel approach to quantify the accuracy of the obtained hosting capacity results is also outlined. Based on the problem formulation, the feeder's PV hosting capacity depends upon two factors: PV deployment scenario (locations and sizes of PV panels), and bus voltages. Bus voltages vary with customer loads and energy output of PV systems; both are varying over time. Therefore, along with modeling the uncertainty in the locations and sizes of the future PV panel deployments, the PV hosting problem is also required to model the impacts of time-series characteristics of load and PV generation on bus voltages. The proposed stochastic analysis framework includes both factors while solving the PV hosting capacity problem.

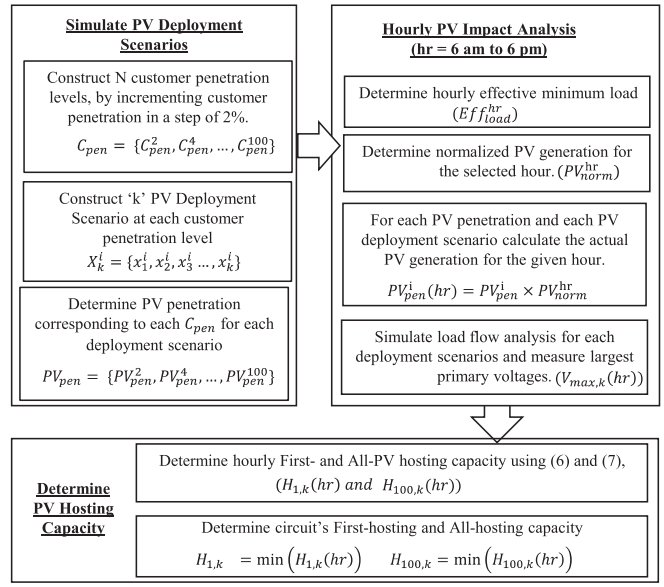

Fig. 1. Proposed PV hosting analysis framework.

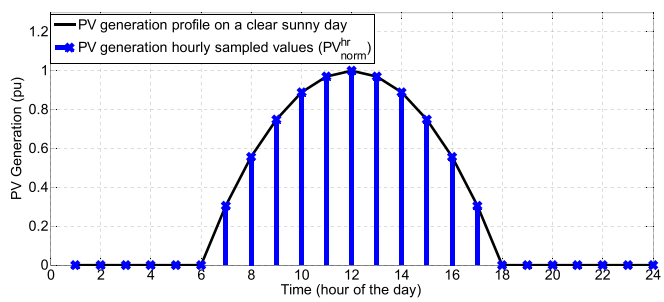

Fig. 2. Typical per-unit PV generation profile on a clear sunny day.

A. Stochastic Analysis Framework

This paper proposes an hourly stochastic analysis framework to solve the PV hosting capacity problem. The proposed methodology can be explained in three steps as outlined in Fig. 1. The first task is to simulate potential PV deployment scenarios. In this work, future PV deployments are characterized by simulating multiple PV deployment scenarios (x_j^i) using the Monte Carlo approach. Next, an hourly PV impact analysis is conducted to analyze the impacts of PV deployment on feeder voltages. The impact analysis requires identifying a representative load and PV generation condition for each hour of the day. Since the paper is concerned with the overvoltage problem, the PV hosting capacity is calculated for the hourly representative minimum load demand obtained using yearly feeder load demand data, termed effective minimum load ($Eff_{\text{load}}^{\text{hr}}$). Similarly, PV generation profile corresponding to a clear sunny day (see Fig. 2) is used to obtain the hourly PV generation for each scenario ($PV_{\text{pen}}^i(\text{hr})$). For each hour, using $Eff_{\text{load}}^{\text{hr}}$ and $PV_{\text{pen}}^i(\text{hr})$, load flow analysis is conducted and the largest primary wire voltages are obtained. Finally, based on the hourly load flow analysis, PV hosting capacities ($H_{1,k}$ and $H_{100,k}$) are calculated (see Fig. 1).

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1) PV Deployment Scenarios: The proposed approach simulates multiple unique PV deployment scenarios by varying PV panel locations and sizes [25]. Using the Monte Carlo approach, k PV deployment scenarios (X_k^i) are simulated at each customer penetration level ($Cust_{\text{pen}}^i$) by associating a uniform random variation to PV panel locations. Additionally, the PV panel size at each customer location is determined based on the customer type (commercial or residential) and the corresponding distributions for the PV panel size [25].

The method to systematically simulate j^{th} PV deployment scenario, where $j \in \{1, 2, \ldots, k\}$, is as follows. First, 2% of customers ($Cust_{\text{pen}}^2$) are selected randomly using a uniform distribution from the pool of customers served by the distribution feeder. At each of the selected customer locations, a PV panel is deployed. The installed PV capacity at each selected customer location is determined based on the customer load type and the corresponding peak load demand. The customer penetration is increased in an increment of 2% and additional PV systems are deployed by selecting locations from the remaining customer loads not connected to PV panels. The customer penetration level is increased and the deployment process is repeated until it reaches 100% ($Cust_{\text{pen}}^{100}$). This constitutes one set of PV deployment scenario corresponding to each customer penetration level. The above process is repeated k times to obtain a total of k PV deployment scenarios (X_k^i) at each i customer penetration level.

2) Hourly PV Impact Analysis: The proposed method evaluates PV impacts in an hourly interval while including the impacts of hourly variations in PV generation and feeder load demand. First, the hourly effective minimum load demand ($Eff_{\text{load}}^{\text{hr}}$) for the feeder is calculated using the yearly net feeder load demand. The measured load demand at the substation includes the contribution of the existing PV generation. The net feeder load demand is obtained after accounting for the existing PV generation in the measured load demand. For each day of the year, the net feeder load demand is sampled at an hourly interval and a percentile analysis is done. For each hour, the 5th-percentile value of the sampled load demand termed $Eff_{\text{load}}^{\text{hr}}$ is obtained. Note that for a given hour, the yearly load demand will be higher than the $Eff_{\text{load}}^{\text{hr}}$ for 95% of the time. Thus, $Eff_{\text{load}}^{\text{hr}}$ characterizes circuit's minimum load demand. The next task is to obtain the hourly PV generation for each PV deployment scenario. The hourly sampled values of the PV generation profile shown in Fig. 2 are used to obtain the hourly active power generation of each PV panel (5).

The impacts of PV generation on feeder voltages are evaluated by simulating a steady-state load flow analysis at each hour for each PV deployment scenario corresponding to each customer penetration level. For a given hour, hr, the load flow analysis is simulated for load demand equal to $Eff_{\text{load}}^{\text{hr}}$ and PV generation equal to $PV_{\text{pen}}^i(\text{hr})$. Using the load flow analysis, the largest primary voltages are recorded corresponding to each PV deployment scenario. Once, the load flow analysis is done, the feeder's first-hosting and all-hosting capacities are calculated corresponding to each hour of the day.

$$PV_{\text{pen}}^i(hr) = PV_{\text{pen}}^i \times PV_{\text{norm}}^{hr} \tag{5}$$

where,

$PV_{\text{pen}}^i(\text{hr})$: PV generation at hour $= hr$ corresponding to i^{th} PV penetration.

$PV_{\text{norm}}^{\text{hr}}$: Normalized PV generation at hour $= hr$.

3) Determine PV Hosting Capacity: Based on the hourly load flow analysis, the first-hosting ($H_{1,k}(hr)$) and all-hosting capacity ($H_{100,k}(hr)$) are determined for each hour using (6) and (7), respectively. Here, $V_{\text{max},k}^i(hr)$ is the set of maximum primary voltages recorded for k PV deployment scenarios simulated for i^{th} customer penetration level at hour $= hr$. Finally, the feeder's first-hosting ($H_{1,k}$) and all-hosting capacity ($H_{100,k}$) are obtained by taking a minimum of the respective hourly values (see Fig. 1).

$$H_{1,k}(hr) = \min_{i \in S} \left\{ PV_{\text{pen}}^i | P(V_{\text{max},k}^i(\text{hr}) > 1.05) \geq \frac{1}{\text{k}} \right\} \tag{6}$$

$$H_{100,k}(hr) = \min_{i \in S} \left\{ PV_{\text{pen}}^i | P(V_{\text{max},k}^i(\text{hr}) > 1.05) = 1 \right\} \tag{7}$$

It should be noted that an hourly load demand lower than the $Eff_{\text{load}}^{\text{hr}}$ is likely to be observed for 5% of the days over a year. Then, ignoring the accuracy of the Monte Carlo simulations, for 95% of days during the year, the actual PV hosting capacity will be at least equal to or greater than the PV hosting capacity calculated using the feeder's effective minimum load demand. Thus, if the total PV generation equal to the calculated hosting capacity is deployed, the circuit is not likely to observe an overvoltage for at least 95% of the days over the year.

B. Accuracy Assessment Framework

A distribution circuit may be supplying for thousands of customer loads. For example, the circuit analyzed in this study is supplying for 1218 customers (see Section IV). Thus, for a 2% customer penetration ($\simeq 25$ customers), the total possible number of PV deployment scenarios will be equal to $\binom{1218}{25} = 6.98 \times 10^{51}$. The Monte Carlo approach, however simulates relatively small number of scenarios. Given the large number of possible PV deployments, it is important to quantify the accuracy of the results obtained using the Monte Carlo method.

The method to define and quantify the percentage accuracy of the hosting capacity results obtained using a k-*run* MCS is presented in this section. The framework is based on the hosting capacity probabilistic intervals identified using the distribution of the hosting capacity results. The pth probabilistic intervals for both first-hosting and all-hosting capacity are defined as the minimum length intervals containing $p\%$ of the corresponding hosting capacity results obtained using multiple k-*run* MCS. Thus, the PV hosting capacity calculated using a k-*run* is expected to be within the pth probabilistic interval for $p\%$ of the time. The accuracy is then defined for the PV hosting probabilistic intervals. It is assumed that if the pth probabilistic interval of the hosting capacity results is accurate, then the hosting capacity obtained using a k-*run* MCS will be accurate for $p\%$ of the time.

1) Impact of Multiple k-run MCS on Hosting Capacity: Given a large number of possible PV deployment scenarios, corresponding to a different k-*run* MCS a different hosting

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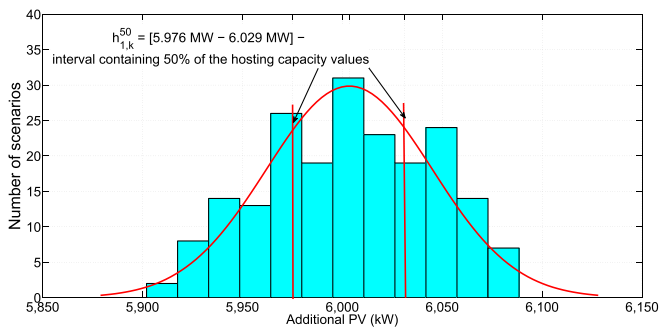

Fig. 3. First-hosting capacity using multiple k-*run* MCS, where $k = 100$.

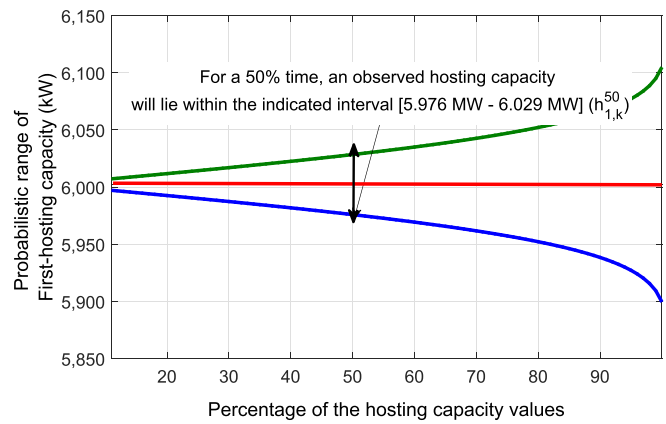

Fig. 4. Probabilistic interval for first-hosting capacity.

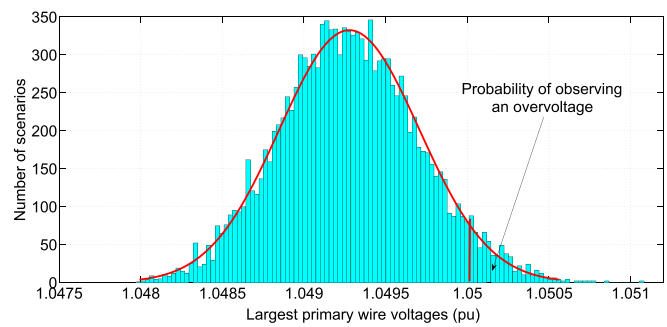

Fig. 5. Largest voltages recorded for 50% hosting capacity interval ($h^{50}_{1,k}$).

capacity is likely to be observed. This observation is illustrated using the first-hosting capacity. We simulated 200 sets of independent k-*run* MCS, where the number of Monte Carlo runs, k, is equal to 100 (100-run MCS). The PV hosting capacity calculated based on the proposed framework resulted in 200 values for first-hosting capacity $(H_{1,k})$. The distribution for $H_{1,k}$ is shown in Fig. 3. Fig. 3 shows that using a 100-run MCS, the first-hosting capacity $(H_{1,k})$ for the feeder can be obtained anywhere from 5.9 MW to 6.1 MW. Since the absolute minimum value cannot be obtained unless we simulate all possible deployments, it is important to determine the accuracy of the results obtained using a single k-*run* MCS.

2) Hosting Capacity Probabilistic Intervals: We begin with obtaining the probability distribution functions (PDF) for both first-hosting $(H_{1,k})$ and all-hosting capacity $(H_{100,k})$. The PDF is then used to obtain the probabilistic intervals for both hosting capacities. The approach to obtain probabilistic intervals is illustrated using $H_{1,k}$.

The PDF (χ^k_{h1}) for $H_{1,k}$ is obtained by approximating the distribution shown in Fig. 3 using a Gaussian distribution. For the obtained PDF (χ^k_{h1}), $\mu = 6.002$ MW and $\sigma = 41.52$ kW.

$$H_{1,k} \sim \chi^k_{h1} = N(\mu, \sigma^2) \tag{8}$$

Using χ^k_{h1}, the pth probabilistic interval for the first-hosting capacity $(h^p_{1,k})$ is calculated using (9).

$$h^p_{1,k} = \left\{ F^{-1}\left(\frac{1-p}{2}|\mu,\sigma\right), F^{-1}\left(\frac{1+p}{2}|\mu,\sigma\right) \right\} \tag{9}$$

Here, $x = F^{-1}(y|\mu,\sigma)$ is defined as $\{x : (F(x|\mu,\sigma) = y)\}$. where,

$$y = F(x|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-(t-\mu)^2}{2\sigma^2}} dt$$

The probabilistic intervals for the first-hosting capacity $(h^p_{1,k})$ calculated for $p = 10\%$ to 99% are shown in Fig. 4.

The distribution of the largest voltages recorded for multiple PV deployments corresponding to each probabilistic interval $(h^p_{1,k})$ is obtained next. The corresponding distribution for $p = 50\%$ is shown in Fig. 5. Note that $h^p_{1,k}$ will be accurate if the probability of observing an overvoltage for the corresponding deployment scenarios is positive but sufficiently small. Also, by definition if $h^p_{1,k}$ is accurate, then $H_{1,k}$ will be accurate for at least $p\%$ of time.

Note that, for a Gaussian distribution, a minimum length p^{th} probabilistic interval is obtained by sampling equal percentage of values on either side of the sample mean. This is possible only because Gaussian distribution is symmetric about the mean. For the cases when the PDF is not symmetric, obtaining the probabilistic intervals calls for a slightly different approach. For completeness, a method for obtaining p^{th} probabilistic interval of a general PDF is detailed in the Appendix Section. Furthermore, in order to validate the analysis presented in this section, the probabilistic intervals are recalculated without making the Gaussian approximation. It is concluded that the probabilistic intervals obtained using Gaussian approximation (see Fig. 4) coincide with the one obtained without making the Gaussian approximation, thus validating the proposed approach. Please refer to the Appendix Section for the detailed discussion.

3) Percentage Accuracy of the Monte Carlo Simulation: For a k-*run* MCS, the percentage accuracy of the feeder's first-hosting and all-hosting capacity is defined using $Acc^{\varepsilon}_k(H_1)$ and $Acc^{\varepsilon}_k(H_{100})$, respectively.

First-hosting capacity $(H_{1,k})$, obtained using a k-*run* MCS, is expected to be accurate for $p\%$ time if the probability of observing an overvoltage for PV deployment scenarios in p^{th} probabilistic interval is greater than zero but sufficiently small, as characterized by the tolerance parameter (ε). $Acc^{\varepsilon}_k(H_1)$ is equal to the maximum p^{th} probabilistic interval satisfying the above definition (10).

All-hosting capacity $(H_{100,k})$, obtained using a k-*run* MCS, is expected to be accurate for $p\%$ time if the probability of

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observing an overvoltage for PV deployment scenarios in p^{th} probabilistic interval is less than or equal to one but sufficiently high, as characterized by the tolerance parameter (ε). $Acc_k^{\varepsilon}(H_{100})$ is equal to the maximum p^{th} probabilistic interval satisfying the above definition (11).

$$Acc_k^{\varepsilon}(H_1) = \max_{0 \le p \le 100} \Big\{ p | 0 < P(V_{\max,k} | h_{1,k}^p > 1.05) \le \varepsilon \Big\} \tag{10}$$

$$Acc_k^{\varepsilon}(H_{100}) = \max_{0 \le p \le 100} \Big\{ p | (1-\varepsilon) < P(V_{\max,k} | h_{100,k}^p > 1.05) \le 1 \Big\} \tag{11}$$

where,

$V_{\max,k}$: Set of largest voltages recorded for each Monte Carlo scenario corresponding to each k-*run* MCS.
$h_{1,k}^p$: p^{th} probabilistic interval for first-hosting capacity.
$h_{100,k}^p$: p^{th} probabilistic interval for all-hosting capacity.
ε: the tolerance parameter.

IV. Results and Discussions

The proposed PV analysis framework is used to determine the PV hosting capacity of an actual 12.47-kV distribution feeder. The results of the PV integration analysis obtained using the proposed framework are compared against those obtained using the statistical minimum load condition [25]. The accuracy of the PV hosting results are calculated as well. Note that the obtained hosting capacity for a distribution circuit depends upon a number of factors, both related to feeder characteristics and simulation method. The paper further evaluates several feeder and simulation related parameters for their impacts on the feeder's PV hosting capacity.

A. Characterizing Distribution Circuit

The selected 12.47-kV distribution circuit is supplied by a 24-MVA substation transformer and is connected to a total of 1.196 MW of PV generation. The distribution circuit serves 1218 customer loads out of which 71% are residential customers. For voltage support, six 600-kvar capacitor banks are distributed along the primary feeder. The three-phase circuit model for the distribution system starting from the substation down to single-phase individual customer load location is simulated. The yearly load demand data measured at the substation for year 2013 is available. For year 2013, the circuit records a peak load demand of 12.5 MW during the summer.

Next, using the yearly load demand measured at the substation for the year 2013, the hourly effective minimum load demand ($Eff_{\text{load}}^{\text{hr}}$) and the statistically representative minimum load demand are obtained for the feeder. Using the method discussed in Section III, the effective minimum load ($Eff_{\text{load}}^{\text{hr}}$) is calculated for each hour of the day (see Fig. 6). The effective minimum load accounts for the existing PV generation. From Fig. 6, the hourly effective minimum load varies from 4.8 MW to 6.8 MW, resulting in a variation of approximately 2 MW. The

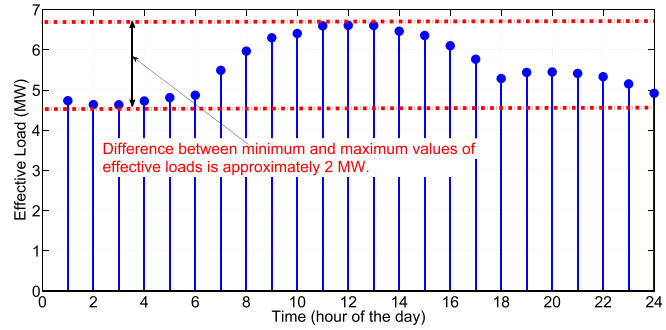

Fig. 6. Hourly effective load calculated for the selected feeder.

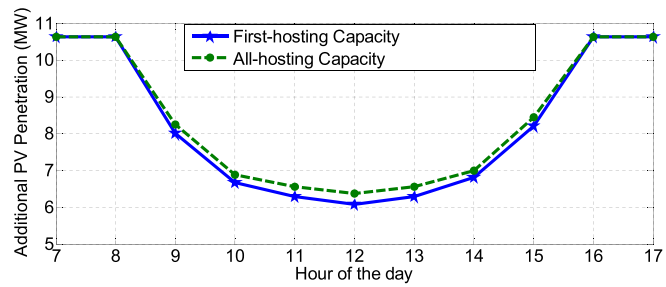

Fig. 7. Hourly PV hosting capacities obtained using proposed framework.

statistically representative minimum load demand is obtained by sampling the daily minimum load demands recorded during daytime hours for the entire year. A histogram analysis is done on the obtained daily minimum load demand values and the mean value of the distribution is obtained. The representative minimum load is selected to be equal to the mean of the distribution, i.e. 6 MW [25].

B. PV Hosting Capacity using the Proposed Framework

Multiple PV deployment scenarios (x_j^i) are simulated using the Monte Carlo approach. The customer penetration level (C_{pen}^i) is varied from 2% to 100% in an increment of 2% and a total of 50 penetration levels are simulated. In this study we have simulated 100 PV deployment scenarios corresponding to each C_{pen}^i. Thus, a total of 5000 cases are simulated for the PV hosting analysis.

Using $Eff_{\text{load}}^{\text{hr}}$, PV hosting capacities are calculated for each hour of the day. The hourly first-hosting and all-hosting capacities are shown in Fig. 7. The minimum of the obtained hourly hosting capacities is taken as the circuit's actual PV hosting capacity. From Fig. 7, the minimum values for both first-hosting capacity and all-hosting capacity are obtained at 12 pm. Therefore, the hosting capacities calculated at 12 pm represent the circuit's actual PV hosting capacity (see Table I). From Table I, the circuit records first violation for primary bus voltages on adding 6.084 MW of additional PV, while all scenarios record overvoltage condition on adding 6.374 MW of additional PV. In between the first- and all-hosting capacities, some scenarios record an overvoltage violation while others do not. The PV hosting results are compared against those obtained for a fixed minimum load condition. Based on the statistical analysis, a

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TABLE I
PV HOSTING RESULTS

Cases		Additional PV Size (kW)
Using Hourly Analysis	First-hosting Capacity ($H_{1,k}$)	6, 084
	All-hosting Capacity ($H_{100,k}$)	6, 374
Using Statistical Minimum Load	First-hosting Capacity ($H_{1,k}$)	5, 454
	All-hosting Capacity ($H_{100,k}$)	5, 722

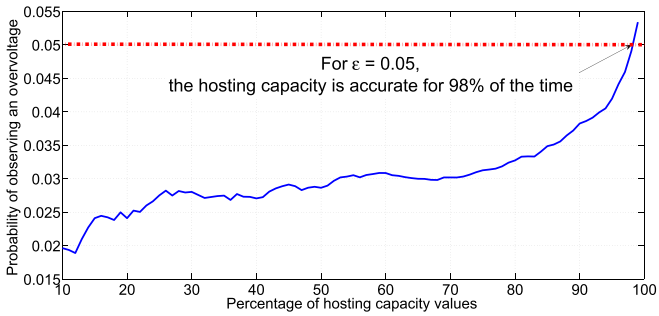

Fig. 8. Probability of observing an overvoltage corresponding to 10% to 99% probabilistic interval for first-hosting capacity.

minimum load equal to 6 MW is selected for the analysis (see Table I). Note that the PV hosting analysis using a fixed load condition underestimates the feeder's PV hosting capacity. This is because the analysis is done at a single minimum load condition and does not include the impacts of the hourly variations in load and PV generation.

Assuming Monte Carlo simulations can capture the representative PV deployments, the first-hosting capacity ($H_{1,k} = 6.084$ MW) is the minimum PV penetration resulting in an overvoltage violation for at least one PV deployment scenario. Since 5th-percentile minimum load is selected, deploying PV generation less than $H_{1,k}$ will not result in an overvoltage for 95% of the days during the year. For 5% of the days when the daily minimum load demand may be lower than the selected load demand, an overvoltage may be recorded on adding an additional PV generation equal to $H_{1,k}$. It should be noted that the interpretation regarding the possibility of overvoltage violation is not possible for the PV hosting capacity obtained using a single load condition. The hourly analysis coupled with percentile load estimation results in a better estimate of the feeder's PV hosting capacity and helps in the statistical interpretation of the PV hosting results.

C. Accuracy of Hosting Capacity Results

The accuracy of the first- and all-hosting capacities is calculated using the accuracy assessment framework detailed in Section III. The tolerance parameter, ε, is assumed to be 0.05. The accuracy is calculated for the stochastic analysis using 100 PV deployment scenarios at each customer penetration level, i.e. a 100-run MCS. Fig. 8 shows the results for first-hosting capacity. The probability of observing an overvoltage for each probabilistic interval of the hosting capacity is obtained and plotted. Using Fig. 8, for $\varepsilon = 0.05$, $Acc_k^{\varepsilon}(H_1) = 98\%$,

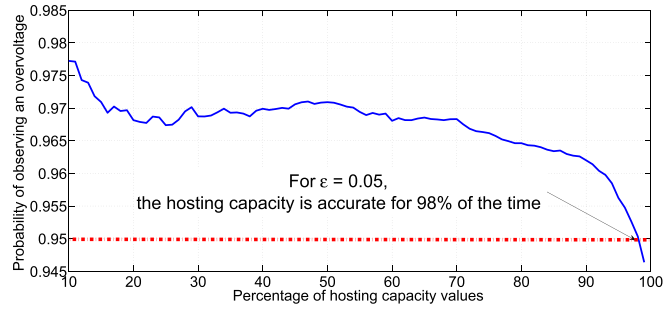

Fig. 9. Probability of observing an overvoltage corresponding to 10% to 99% probabilistic interval for all-hosting capacity.

implying that the first-hosting capacity obtained using a 100-run MCS is expected to be accurate for 98% of the time over the year. Similarly, the accuracy of all-hosting capacity is obtained (Fig. 9). From the figure, for $\varepsilon = 0.05$, the all-hosting capacity calculated using an independent 100-run MCS is expected to be accurate for 98% of the time.

D. Impacts of Distribution Circuit Characteristics on PV Hosting Capacity

The obtained PV hosting capacity for a given circuit is sensitive to the distribution circuit characteristics. This section details the impacts of several circuit parameters on the selected circuit's PV hosting capacity. The impacts of some of the circuit characteristics such as loads and PV panel locations were evaluated in [25]. This paper includes the impacts of several additional parameters including substation short-circuit capacity, line R/X ratio, and voltage regulator setting. It is observed that the distribution feeder with a higher short-circuit capacity and a lower R/X ratio can accommodate a higher penetration of PV generation. Also, as the voltage regulator setting for the voltage of the substation bus decreases, the circuit's PV hosting capacity increases.

1) Substation Short-Circuit Capacity: The substation short-circuit capacity for the selected distribution circuit is 160 MVA. The short-circuit capacity of the substation is increased from 60 MVA to 240 MVA. Table II shows the first- and all-hosting capacities obtained for the circuit corresponding to each value of the substation short-circuit capacity. It can be observed that as the short-circuit capacity increases the PV hosting capacity of the circuit also increases.

2) Line Impedance - R/X Ratio: In this section, the impacts of the distribution circuit's line impedance are evaluated on the feeder's PV hosting capacity. Note that the selected circuit is an actual utility feeder supplied by different types of primary and secondary distribution lines with a range of R/X characteristics. For the given circuit, the average R/X ratio of the secondary lines is 0.25. For this section, first all secondary lines are replaced with a common distribution line type of a specified R/X ratio. The lines' R/X ratios are then varied from 0.25 to 5 and the PV hosting capacity of the circuit is calculated. The results are detailed in Table II. As expected the circuit's PV hosting capacity significantly decreases on increasing the R/X ratio of the secondary lines.

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TABLE II
IMPACTS OF DISTRIBUTION CIRCUIT CHARACTERISTICS ON PV HOSTING CAPACITY

Substation short circuit capacity (SCC)			Secondary line's R/X ratio			Voltage regulator (Vreg) setting		
SCC (MVA)	$H_{1,k}$ (MW)	$H_{100,k}$ (MW)	R/X	$H_{1,k}$ (MW)	$H_{100,k}$ (MW)	Vreg (pu.)	$H_{1,k}$ (MW)	$H_{100,k}$ (MW)
60	4.79	5.21	0.25	6.02	6.28	0.99	10.64	10.64
100	5.53	5.95	0.5	5.44	6.03	1.00	10.64	10.64
140	5.89	6.28	2	3.72	4.61	1.01	10.64	10.64
160	6.08	6.37	3	3.04	4.09	1.02	9.41	9.56
200	6.22	6.46	4	2.75	3.70	1.03	6.08	6.37
240	6.29	6.55	5	2.41	3.48	1.04	4.22	4.54

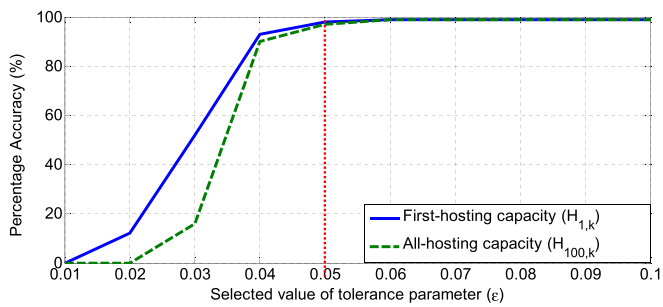

Fig. 10. Hosting capacity percentage accuracy vs. tolerance parameter for $k = 100$.

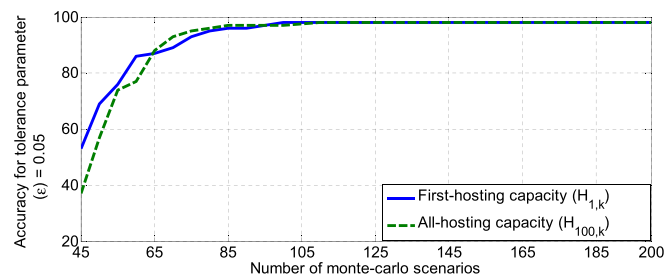

Fig. 11. Hosting capacity accuracy vs. the number of Monte Carlo runs.

3) Voltage Regulation Equipment Setting: The impacts of substation regulator's voltage setting on the PV hosting capacity are evaluated. The substation bus voltage that the voltage regulator is required to maintain is changed from 0.99 pu to 1.04 pu. The PV hosting analysis is done and results are shown in Table II. From the table, the circuit can accommodate more PV generation when the substation bus is maintained at a lower voltage. This is because the voltage rise due to PV is more likely to result in an overvoltage limit violation if the feeder voltages are already high.

E. Impacts of Simulation Parameters on PV hosting Capacity

The PV hosting results for a given distribution circuit are also affected by simulation parameters. Therefore, it is important to observe the impacts of various simulation parameters such as tolerance parameter (ε) and number of Monte Carlo runs on the feeder's PV hosting results.

1) Impact of Tolerance Parameter: The impact of the tolerance parameter (ε) on the accuracy of hosting capacities is analyzed next. Fig. 10 shows the percentage accuracy of first- and all-hosting capacities with respect to ε. As expected, as the tolerance is increased, the accuracy of hosting capacity increases. Also, for $\varepsilon = 0.06$, both $Acc_k^{\varepsilon}(H_1)$ and $Acc_k^{\varepsilon}(H_{100})$ are greater than 99%.

2) Impact of the Number of Monte Carlo Runs: The PV hosting capacity accuracy also depends upon the number of Monte Carlo runs/scenarios. Increasing the number of Monte Carlo runs can better approximate the PV deployment scenarios and increase the confidence in the obtained hosting capacity values.

In this section, the PV hosting capacity is calculated for different numbers of Monte Carlo runs with ε equal to 0.05. The number of Monte Carlo runs is varied from 45 to 200 at each customer penetration level, and the percentage accuracy of both first- and all-hosting capacity are obtained. Fig. 11 shows the relation between hosting capacity accuracy and the number of Monte Carlo runs. It can be observed from the figure that although the accuracy increases with the increase in the number of Monte Carlo runs, a reasonable accuracy is obtained on simulating less than 100 Monte Carlo scenarios. In fact, more than 90% accuracy is obtained just by simulating 75 Monte Carlo runs at each customer penetration level.

V. CONCLUSION

This paper presents a framework to understand and solve the PV hosting capacity problem for utility distribution circuits. First, a mathematical formulation for the hosting capacity problem is developed. To solve the formulated problem, an hourly stochastic analysis approach is proposed. The method is illustrated by calculating the hosting capacity of an actual 12.47-kV feeder for overvoltage concerns. The results obtained using the proposed method are compared against those obtained using a fixed minimum load condition. It is concluded that the proposed approach results in a better estimate of the feeder's PV hosting capacity. Next, an approach to evaluating the percentage accuracy of the hosting capacity results is developed. The method aims to identify the percentage of time a hosting capacity obtained using a *k-run* MCS is expected to be accurate. The percentage accuracy is determined with respect to a tolerance parameter (ε). For $\varepsilon = 0.05$, both first- and all-hosting capacities calculated using a 100-run MCS are concluded to be accurate for 98% of the time.

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Additionally, the paper evaluates the impacts of distribution circuit characteristics and simulation parameters on hosting capacity results. It is observed that the circuit can accommodate additional PV generation if the substation is strong or the R/X ratio of the secondary lines is low. The feeder's PV hosting capacity increases on decreasing the per-unit value of the substation regulator's voltage setting. It is also observed that the accuracy of the obtained PV hosting capacity increases both on decreasing the tolerance parameter and on increasing the number of Monte Carlo runs. However, a reasonable accuracy (>90%) is obtained on simulating 75 Monte Carlo runs and greater than 99% accuracy is guaranteed by simulating only 105 runs.

Although the results are demonstrated using overvoltage condition, the proposed framework can be applied to determine the hosting capacity for other impact criterion, such as voltage deviation, voltage imbalance, etc. Additionally, the proposed framework can be used to include the impacts of PV variability as well. To do so, the seasonal and location dependent PV data can be statistically analyzed to obtain a representative hourly PV generation profile at each PV location. The obtained hourly PV profiles can be used instead of the clear sky PV profile in the proposed analysis framework. In conclusion, by developing a mathematical formulation, an hourly stochastic analysis framework, an approach to quantify the percentage accuracy of the results, and by performing the sensitivity analyses this paper presents a comprehensive understanding of the distribution feeder's PV hosting capacity problem.

APPENDIX

The objective of this section is to validate the credibility of the probabilistic intervals calculated in Section III-B. In Section III-B, the probabilistic intervals were obtained by approximating the hosting capacity results using a Gaussian distribution (see Fig. 3). In this section, an empirical PDF is obtained for the hosting capacity results and the probabilistic intervals are recalculated without using the Gaussian approximation. First, we present a simple heuristic to calculate the probabilistic intervals of a general or an empirically obtained PDF (see Algorithm 1). The proposed heuristic is used to determine the probabilistic intervals for the empirically obtained distribution of the first-hosting capacity. It is observed that the probabilistic intervals obtained with Gaussian approximation coincide with those obtained without making the Gaussian approximation. Therefore, in this case, the Gaussian approximation is valid for the accuracy assessment. Furthermore, Algorithm 1 can be used to determine probabilistic intervals and the accuracy of the hosting capacity results when the observations do not follow a Gaussian distribution, thus broadening the scope of the work.

A. Probabilistic Intervals of a General Probability Distribution Function

Recall that the pth probabilistic intervals for both first-hosting and all-hosting capacity are defined as the minimum length intervals containing $p\%$ of the corresponding hosting capacity results obtained using multiple *k-run* MCS. Using the definition,

Algorithm 1: Calculate p^{th} probabilistic interval, $h^{p}_{1,\,k}$, for First-hosting capacity $H_{1,\,k}$.

Require: PDF χ^{k}_{h1} corresponding to $H_{1,\,k}$.
Define $K = \{x : 0 < x < 1 \text{ and } 100x \text{ is an integer}\}$
Define $(k_1, k_2) = \{(x, x+p)\text{: } x \in K \text{ and } (x+p) < 1\}$
Define $n = N(k_1, k_2)$, where, $N(A)$ is equal to the number elements in the set A
Calculate CDF, F, corresponding to $H_{1,\,k}$ using χ^{k}_{h1}
for all $(k_1(i), k_2(i)) \in (k_1, k_2)$ **do**
  Calculate $h^{p}_{1,\,k}(i) = \left\{F^{-1}\left(k_1(i)\right), F^{-1}(k_2(i))\right\}$.
end for
Select $(k_1(opt), k_2(opt))$ with minimum value of $\left\{F^{-1}\left(k_2(i)\right) - F^{-1}(k_1(i))\right\} \forall i \in n$.
return $h^{p}_{1,\,k} = \left\{F^{-1}\left(k_1(opt)\right), F^{-1}(k_2(opt))\right\}$.

the pth probabilistic interval $h^{p}_{1,\,k}$ is defined as the following.

$$h^{p}_{1,\,k} = \left\{F^{-1}\left(k_1\right), F^{-1}\left(k_2\right)\right\} \tag{12}$$

where, $k_2 - k_1 = p$, and F is the cumulative distribution function (CDF) for the selected hosting capacity ($H_{1,\,k}$ or $H_{100,\,k}$).

Note that, for a given p, k_1 and k_2 can hold multiple values. For example, for $p = 0.5$, (k_1, k_2) can be selected from the following set: $\{(0.1, 0.6), (0.2, 0.7), (0.25, 0.75), ...\}$. Depending upon the pdf, for each set of (k_1, k_2) corresponding to a given p, $h^{p}_{1,\,k}$ of different lengths will be obtained. This case is shown in Fig. 12.

The objective is to obtain (k_1, k_2) such that the probabilistic interval, $h^{p}_{1,\,k}$, is of the minimum length. For a symmetric PDF, a minimum length $h^{p}_{1,\,k}$ is obtained by sampling equal percentage of values on either side of the sample mean. Therefore, for the PDF approximated using Gaussian distribution, k_1 and k_2 are equal to $\frac{1-p}{2}$ and $\frac{1+p}{2}$, respectively (see (9)). However, for a general distribution, we need to obtain k_1 and k_2 resulting in the minimum length p^{th} probabilistic interval.

Therefore, the pth probabilistic interval of the cumulative distribution function (CDF) defined as: $y = F(x)$, is obtained by calculating k_1 and k_2 using (13).

$$\min.\left\{F^{-1}\left(k_2\right) - F^{-1}\left(k_1\right)\right\} \tag{13}$$

such that

$$k_2 - k_1 = p$$

A simple heuristic for obtaining probabilistic intervals of a general distribution, corresponding to the problem defined in (13), is detailed in the following section.

B. Heuristic Proposed for Obtaining Probabilistic Intervals

The approach for obtaining p^{th} probabilistic interval, $h^{p}_{1,\,k}$, for first-hosting capacity ($H_{1,\,k}$) with PDF χ^{k}_{h1} is detailed in Algorithm 1.

The algorithm begins with enumerating possible values for (k_1, k_2) corresponding to a given value of p. Note that,(k_1, k_2) can take infinite values for a given p. In the proposed heuristic,

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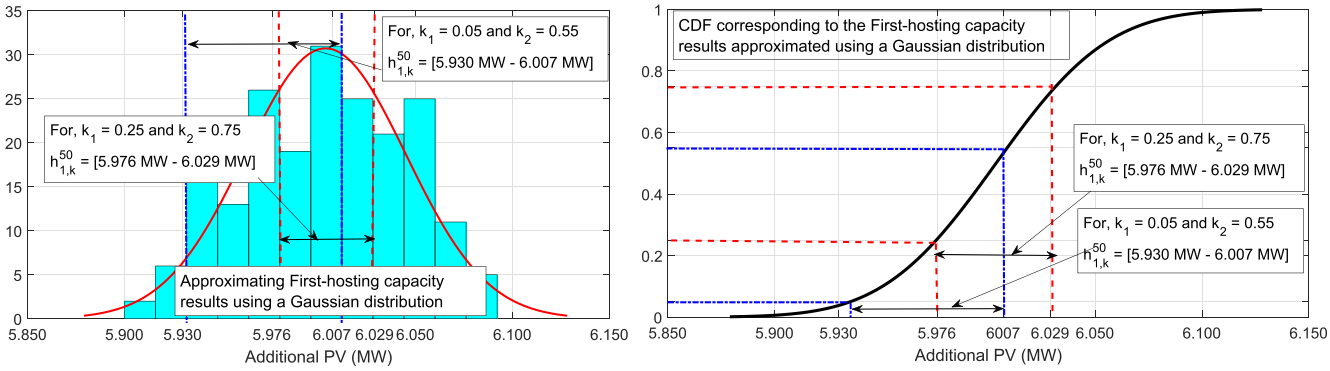

Fig. 12. Probabilistic intervals explained using cumulative distribution function (CDF). The length of pth probabilistic interval ($h^{p}_{1,\,k}$) changes on modifying (k_1, k_2). For $(k_1, k_2) = (0.25, 0.75)$, the obtained p^{th} probabilistic interval, $h^{p}_{1,\,k} = (5.976, 6.029)$ is of smaller length than the one obtained for $(k_1, k_2) = (0.05, 0.55)$. Note that for both case $p = (k_2 - k_1) = 0.5$.

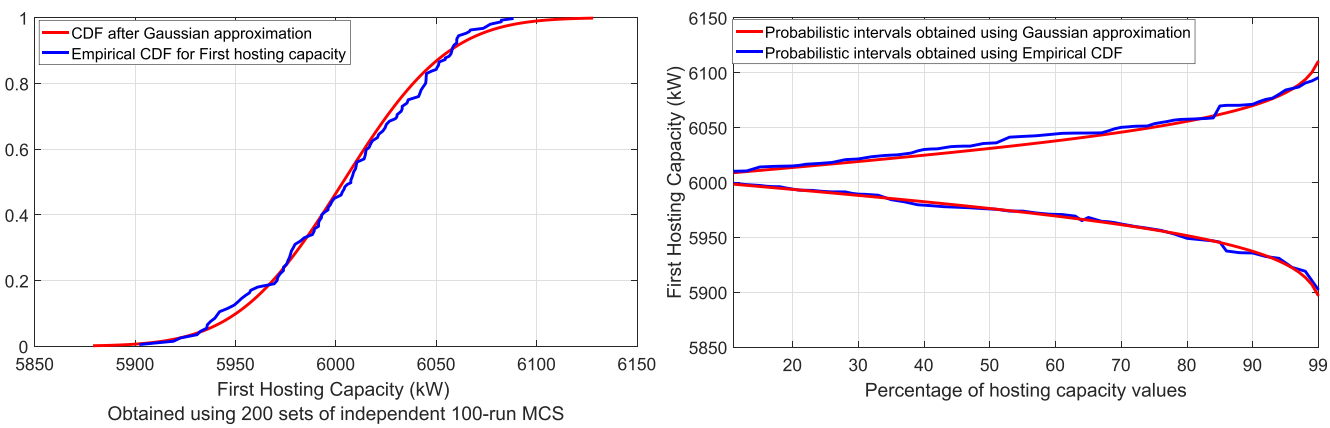

Fig. 13. Comparing the cumulative distribution function (CDF) and probabilistic intervals obtained using a Gaussian approximation and using empirical methods; a) CDF for first-hosting capacity ($H_{1,\,k}$), b) Probabilistic intervals obtained for first-hosting capacity ($H_{1,\,k}$).

we enumerate (k_1, k_2) over a finite set $K = \{x : 0 < x < 1$ and $100x$ is an integer$\}$. Next, we calculate the pth probabilistic interval, $h^{p}_{1,\,k}$, for each (k_1, k_2). Finally, the minimum length $h^{p}_{1,\,k}$ is selected.

C. Probabilistic Intervals Obtained Without Gaussian Approximation

In this section, we demonstrate the validity of approximating hosting capacity results using a Gaussian distribution in the proposed accuracy assessment framework. Recall that in Section III-B, probabilistic intervals were obtained by approximating first- and all-hosting capacities using a Gaussian distribution. In this section, first, we determine the actual distribution for the first-hosting capacity ($H_{1,\,k}$) using empirical methods. Both empirically calculated CDF and the CDF obtained using the Gaussian approximation for $H_{1,\,k}$ are shown in Fig. 13(a) . It is observed from the figure that empirically obtained distribution closely follows the Gaussian distribution. Next, we obtain probabilistic intervals ($h^{p}_{1,\,k}$) using the empirically obtained CDF for $H_{1,\,k}$. The probabilistic intervals obtained using the empirically calculated distribution also coincides with those obtained using a Gaussian approximation (see Fig. 13(b)). Thus, the Gaussian approximation is valid for the hosting capacity accuracy assessment of the selected feeder.

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Anamika Dubey (M'16) received the M.S.E and Ph.D. degrees in electrical and computer engineering from the University of Texas at Austin, Austin, TX, USA, in 2012 and 2015, respectively. She is currently an Assistant Professor in the School of Electrical Engineering and Computer Science, Washington State University, Pullman, WA, USA. Her research interests include distribution system analysis and planning, and grid integration of distributed energy resources, and electric vehicles.

Surya Santoso (F'15) received the M.S.E. and Ph.D. degrees from the University of Texas at Austin, Austin, TX, USA, in 1994 and 1996, respectively. He has been a Professor at the University of Texas at Austin since 2003. He has co-authored the book entitled *Electrical Power Systems Quality-Third Edition* (New York, NY, USA: McGraw-Hill) and authored the book entitled *Fundamental of Electric Power Quality* (2013). His research interests lie in the broad area of electric power quality and wind power integration.

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